COMPUTATIONAL ELECTRODYNAMICS

# Method of Images Notes

This is the version I want in my head when solving problems: not just the image-charge formulas, but why the construction is legal, when it works, and what to check before trusting it.

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**TOPIC**
Method of images for grounded conductors

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## 1. What the method is actually doing

In electrostatics the potential satisfies Poisson's equation in source regions and Laplace's equation in charge-free regions.

```
Laplace(phi(r)) = -rho(r) / eps0
```

```
Laplace(phi(r)) = 0   in source-free regions
```

The hard part is usually the boundary condition on the conductor. The method of images replaces that boundary by fictitious charges outside the physical region, chosen so that the candidate potential automatically satisfies the boundary.

1. Solve the right PDE in the physical region.
2. Satisfy the boundary exactly.
3. Keep image singularities outside the physical region.
4. Use uniqueness.

## 2. Why uniqueness makes the method rigorous

For a Dirichlet problem, if the charge density in the region is specified, the boundary potential is specified, and the behavior at infinity is fixed when needed, then the electrostatic solution is unique.

That means the image charges do not need to be physically present. They only need to help construct a potential that solves the same problem in the physical region.

## 3. Checklist before trying images

- Is the geometry symmetric enough to admit an exact image construction?
- Is the conductor grounded or isolated?
- What is the physical region?

- Where are the real charges?
- Can all images be placed outside the physical region?
- Can the boundary be verified exactly?

## 4. Grounded infinite plane

Take a point charge q at (0,0,a) with a > 0 above the grounded plane z = 0. The physical region is z > 0.

```
real charge:  q  at (0,0,a)
image charge: -q at (0,0,-a)
```

```
phi(r) =
(1 / 4 pi eps0)
[ q / |r-r0| - q / |r-r0*| ],   z > 0
```

On the plane the distances to the real and image charge are equal, so phi = 0 exactly. The only singularity in z > 0 is the real charge, and the potential decays correctly at infinity. By uniqueness, the construction is valid.

Main anchors here: the method-of-images notebook and formula sheet.

## 5. Field, induced charge, and force for the plane

The induced surface charge density comes from the normal field just above the conductor.

```
sigma(rho) = eps0 Ez(rho,0+)
           = -eps0 (dphi/dz)|_{z=0+}
```

```
sigma(rho) = - q a / [ 2 pi (rho^2 + a^2)^(3/2) ]
```

This result is negative everywhere for a positive external charge and is strongest directly below the charge.

```
Q_ind = integral sigma(rho) 2 pi rho d rho = -q
```

The force on the real charge is the Coulomb force due to the image charge:

```
F = - q^2 / (16 pi eps0 a^2)  z-hat
```

The charge is attracted toward the plane. The shortcut is legitimate because the image reproduces the correct field in the physical region, not because there is literally a charge behind the plane.

## 6. Energy subtlety for the plane

The force can be computed directly from the image, but the energy is not the full Coulomb interaction energy between the real and image charge treated as two independent charges.

```
U = (1/2) q phi_image(r0)
```

The factor of one-half matters because the conductor response is induced.

## 7. Grounded conducting sphere

Now place a real charge q on the z-axis a distance a from the center of a grounded sphere of radius R, with a > R.

```
q' = -q R/a
b  = R^2/a
```

```
phi(r) =
(1 / 4 pi eps0)
[ q / |r-a z-hat| + q' / |r-b z-hat| ],   r > R
```

The image sits inside the sphere, outside the physical exterior region. The curved boundary means the image changes both position and magnitude, unlike the plane case.

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Main anchors here: grounded-sphere derivations in the notebook and the sphere GIF code.

## 8. Induced charge and force for the sphere

The surface charge density comes from the outward radial derivative at the surface:

```
sigma(theta) = -eps0 (dphi/dr)|_{r=R+}
```

```
sigma(theta) =
- (q / 4 pi R)
  (a^2 - R^2) / (a^2 + R^2 - 2 a R cos(theta))^(3/2)
```

The induced charge is negative everywhere and is most concentrated near the point closest to the external charge.

```
Q_ind = q' = -q R/a
```

```
F = - (1 / 4 pi eps0) q^2 R a / (a^2 - R^2)^2  z-hat
```

Two checks are worth keeping in mind: as a goes to infinity, the force goes to zero; as a approaches R from above, the attraction grows strongly. In the local flat-surface limit, the sphere result approaches the plane result.

## 9. Plane versus sphere

**Shared structure**

- both are grounded Dirichlet problems
- both use images outside the physical region
- both rely on exact boundary cancellation
- both are justified by uniqueness

**Important differences**

- plane: simple mirror image with opposite sign
- sphere: image changes both position and magnitude
- plane: total induced charge is -q
- sphere: total induced charge is -qR/a

## 10. Relation to Green's functions

```
G_D(r,r') = 1/|r-r'| - 1/|r-r'*|
```

For the grounded plane, the method of images can be read as constructing a Dirichlet Green function for the half-space. So the method is not separate from Green-function theory. It is a special exact case of it.

## 11. Grounded versus isolated conductors

A grounded conductor has fixed potential and can exchange charge with ground. An isolated neutral conductor is a different problem: the total charge is constrained and the image construction generally changes.

## 12. When the method fails

- irregular geometries
- insufficient symmetry
- cases where only an approximate boundary match is possible

In those cases, use separation of variables, Green functions in a more general form, or numerical methods.

## 13. What I actually want to remember

- The method is justified by uniqueness, not by reality of the images.
- Images must stay outside the physical region.
- Grounded plane: mirror charge -q.
- Plane induced charge density: -q a / [2 pi (rho^2 + a^2)^(3/2)].
- Grounded sphere: q' = -qR/a and b = R^2/a.
- The method of images is really a special Green-function construction.
- Grounded and isolated conductors are different problems.

Sources: Jackson PDF, the method-of-images notebook, formula sheet, problem set, and the grounded-sphere visualization code in this repo.